

Christopher Barker

Frontend Web Developer / UX/UI / Data Visualization
Tacoma, WA • Christopher.Barker@plu.edu • LinkedIn • GitHub

Education

- **Pacific Lutheran University** — Bachelor of Arts, Individualized Major in Web Development and Interactive Design (2024–2026)
- **Tacoma Community College** — Associate of Arts, Direct Transfer Agreement (2017–2022)

Technical Projects

- **AI Protein Sequence Visualization & Annotation** — Built a multimodal AI pipeline using ESM-3 and BLIP-2 to convert protein sequences into visual embeddings and generate protein function captions.
- **Derailed** — Developed a multiplayer browser game using React and JavaScript with responsive gameplay interfaces and real-time collision logic.
- **West Coast Tourism Data Analysis** — Engineered Python workflows analyzing spending behavior, tourism trends, and basket diversity metrics.
- **Image-Aware Caption Generator with Vibe Control** — Developed an AI-powered caption generator using Streamlit and OpenAI APIs for image-aware social media captioning.
- **Discord Reminder Bot** — Built a Python-based reminder bot featuring slash commands, timers, and modular workflows.

Work & Research Experience

- **Undergraduate Research Assistant — Pacific Lutheran University**
Authored Python scripts for biological sequence analysis and visualization workflows supporting interdisciplinary AI research.
- **PC Technician — Pacific Lutheran University**
Troubleshoot hardware/software issues, managed inventory workflows, and mentored student workers.
- **UX/UI Design Intern — TestUnlimited.com**
Improved product page usability, created wireframes, and collaborated on front-end design implementation.

Achievements & Awards

- Consortium for Computing Sciences in Colleges (CCSC) — Northwestern Conference (2025)
- Dean's List — University of Washington Tacoma (2024–Current)
- First Generation Fellow — University of Washington Tacoma (2022–2023)
- Outstanding Student Award — Tacoma Community College (2022–2023)

Technical Skills

- **Languages:** JavaScript, Python, Java, HTML5, CSS3, JSON, R
- **Frameworks & Libraries:** React, Material UI, Node.js, JUnit, Streamlit

- **Tools & Platforms:** Git, GitHub, Docker, Figma, VS Code, IntelliJ, RStudio